

I'm Counting On It

Introduction

Alas, a book that can be judged by its cover! The goal here is going to be to introduce you, my dear reader, to actual mathematics through a series of counting games. There isn't much mathematical maturity expected here, but I am expecting a sufficiently weighted balance of curiosity, perseverance, and a capacity to reason at least around the high school math level. Don't be afraid to use a search engine, reach out to a friend, ask a teacher, or heaven forbid go to a library if you get stuck; research skills are as much, if not more important than any other skill in the mathematician's toolbox. Seriously, look stuff up. In academia proper, Wikipedia is generally frowned upon, but it has been a wonderful resource in my mathematics education thus far.

As in any field of study, math has **jargon**. I try to be as kind to the reader as possible and define important terms as they'll be used, as well as giving a computer science student's perspective.

Something important to explicitly discuss up front is the importance of letters in mathematics. Using placeholders like letters instead of numbers lets us instead focus on the logic surrounding the specific values we can operate on. For that focus on logic, I'll rely heavily on the use of letters in this text. I will try to keep some class about it and use contextual letters when thinking of a problem in a certain way, ie. l, w, h when thinking in geometric dimensions, s or t when thinking sequentially, i for indexing, you get the gist physicist. Number theory and combinatorics texts I've seen tend to use k, n, m, p , and even just the letters in alphabetical order, I default to this often when there's little rhyme or reason.

Acknowledgements

First and foremost a thank you to my parents; your unconditional love, support, and faith in my direction has given me the ability to follow my curiosity. Grant Sanderson and his wonderful Youtube Channel 3Blue1Brown deserve academy awards level praise for Grant's ability to elucidate and communicate mathematics. A thank you must go to Sal Khan and the Khan Academy team for helping me shape up for my first few college math courses and set myself on this journey. Richard Hammack and his free online "Books of Proof" have been indispensable in teaching me the basic language and handing me a few important tools of the trade.

An offering to the Pythagoreans and René Descartes.

Setting The Stage

I will not lie to you big dawg it is about to get wicked. Set theory, as the basis for most of modern mathematics, has earned itself a full chapter upfront. This is not material that must be mastered to progress, in fact I urge you, dear reader, to skip ahead and start absorbing as many counting techniques as you can, this chapter'll still be here. Some of the ideas presented here may feel very abstract and disconnected from reality, but this language of set theory will help you read many technical books and papers.

Mathematicians do most of their work with sets, so that's where we'll begin. In general, a set is a collection of objects (elements) with no repetition or order. We will usually denote sets with bold capital letters, S , and define them in curly braces.

For an example, $D = \{\bullet, \bullet, \bullet, \bullet, \bullet, \bullet\}$ is a set of six dots, each colored differently. Notice that by our definition, if we have the set $P = \{\bullet, \bullet, \bullet, \bullet, \bullet, \bullet\}$, $P = D$, but $V = \{\bullet, \bullet, \bullet, \bullet, \bullet, \bullet\}$ is not a set since the green dot repeats.

Stick with me because there are a handful more symbols and a few logical rules that'll really help us say mathematical things with relative cleanliness.

Important Sets and Notation

Many Sets are important enough to the study of mathematics that we use special fonts and (relatively) permanent. I'm noticing a pattern at least insofar as the Number Systems usually using a font with extra lines. Some important sets for us include the **Empty Set** $\emptyset = \{\}$, which is pretty self explanatory, the **Natural Numbers** $\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$ are zero and the standard counting numbers. Some would consider zero to not be apart of the naturals and would tell us that above we have (sloppily) defined the set of whole numbers. See von Neumann's construction using the empty set buddy. $\mathbb{Z} = \{0, 1, -1, 2, -2, 3, 4, -4, \dots\}$ is the set of **Integers**.

Set builder notation (SBN) is how we read and write in the language of sets, and what follows will be an introduction to that language. When defining a set, we may find it helpful to read the opening bracket, "{", as "The set of all objects of the form..."; this is usually followed by a colon ":" (some authors use the pipe "|"), which you read as "Such that."

Therefore if I want to write in **SBN** "E is the set of all even numbers," I say " $E = \{n : n \text{ is even}\}$."

Since Sets are studied from a mathematical POV, it's customary to use any sufficient and necessary mathematical symbols in the **Statements** we use to construct and talk about sets. Alongside mathematics, **Symbolic logic** is a key subject from which we adopt many of our hieroglyphs. Symbolic logic systems encode statements and propositions into variables (it's customary to use $p, q,$ and r , but any agreed upon symbol works) and express logical operations using symbols.

The **universal quantifier**, " $\forall$ ", is short-hand notation for "for all" or, "for everything of the form". Our second quantifier, the **existential quantifier**, " $\exists$ " means "there exists". We can now write statements like "every natural number, n , has a square, m , that is also natural," as " $\forall n \in \mathbb{N}, \exists m \in \mathbb{N} : m = n^2$ ".

Equally as important as propositional variables and quantifiers are **logical connectives**. Connectives help us use "if-then," "and," "or," and "not" to determine the truth of mathematical and logical statements. We say "the statement p implies the statement q ," as " $p \Rightarrow q$ ". The phrase "the statements r and s ," is written " $r \wedge s$," and or just looks like a capital letter v , so " p or s " is " $p \vee s$ ". The symbol for negation is sometimes written as a tilde " $\sim$," and sometimes the symbol " $\neg$," so "not q " is written either " $\sim q$ " or " $\neg q$ ".



Membership, Subsets, and Cardinality

The Symbol for Set membership, $\in$, which may be expressed as "is/are (an) element(s) of," kind of looks to me like an E or epsilon. The English statement "negative Sixty-Seven is a member of the Set of integers" can be written Set theoretically as " $-67 \in \mathbb{Z}$." The next concept is that of the subset, denoted with a symbol that looks like a less than or equal to inequality sign with the closed end being rounded rather than acutely angled; the english phrase "The set of natural numbers is a subset of the set of integers" can be expressed as " $\mathbb{N} \subseteq \mathbb{Z}$." Note that a set is a subset of itself.

Not all sets have the same number of elements, and there are cases in which that "number of elements," formally called **cardinality**, is important to us. Cardinality is denoted with the vertical bar symbol used as a grouping symbol like parentheses, therefore if:

$$D = \{ \bullet, \bullet, \bullet, \bullet, \bullet, \bullet \} \text{ then } |D| = 6, \text{ and if } A = \{ a, b, c \}, \text{ then } |A| = 3$$

If $S \subseteq U$, and $|S| < |U|$, then we say that S is a **proper subset** of U , and write it $S \subset U$.

We can now say that " $X = \{\bullet, \bullet, \bullet\} \Rightarrow X \subset D$ ", and interpret that if we define a set X whose elements are all black, blue, and red dots, then we know X is a proper subset of set D . And we know this to follow through from our definitions so far.

A special case we may consider is the set of all subsets of our example set X . This set of subsets is called the power set of X , is denoted:

$$P(X) = \{\{\}, \{\bullet\}, \{\bullet\}, \{\bullet\}, \{\bullet, \bullet\}, \{\bullet, \bullet\}, \{\bullet, \bullet\}, \{\bullet, \bullet, \bullet\}\}$$

One thing you can convince yourself by ordering a power set like we have above is that for every subset S which exists as a member of a set X , there exists another subset of X whose elements are exactly all of those in X , but not in S . We'll call this curious set the complement of S and denote it as $\bar{S}$.

$$\bar{S} = \{x: x \in X \wedge x \notin S\}$$

Set operations and Indexing

The big three Set operations we'll worry about are going to be Union, difference, and intersection.

I like to think of unions as a kind of careful addition.

Unions help us determine objects that are elements of any one of a number of given sets. The symbol for a set union looks like a big "u", $\cup$. The union is defined as $X \cup Y = \{x : x \in X \vee x \in Y\}$.

Now let's look at the union of the Example sets

$$D \cup A = \{\bullet, \bullet, \bullet, \bullet, \bullet, \bullet, a, b, c\}$$

The difference between sets is given by the minus sign and is defined as $U - S = \{x : x \in U \wedge x \notin S\}$

You've got a green light for set intersections. Intersections are denoted with the union symbol upside down and is defined $X \cap Y = \{x : x \in X \wedge x \in Y\}$

Venn diagrams were introduced by mathematician John Venn in the first article in the July 1880 issue of The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science.

In this essay, an early quote of Venn's reads

The great bulk of the propositions which we commonly meet with are founded, and rightly founded, on an imperfect knowledge of the actual mutual relations of the implied classes to one another.

Sequences, Sums, Series

Oh my! You, my dear reader, are already intuitively familiar with at least one of these concepts, but by the end of this journey will be intimately familiar with all three and the roles they play in the grand theatre of discrete mathematics. If you've taken calculus II, you should be familiar with how important they are for communicating and solving problems on the continuous side of the mathematical coin.

A **Sequence** is an infinite list $a_1, a_2, a_3, a_4, \dots$ usually denoted by mathematicians with the notation $\{a_s\}$, where position is given by the index s .

A list is a collection of objects in which repetition is allowed and order defines unique identity.

A **Sum** is simply an addition; more fondly, a sum is the first combinatorial step taken past counting on our fingers. If you leave here with any one idea lodged inbetween your ear-holes, I hope that idea is that there are seemingly countless counting techniques.

A **Series** is then the love-child of the preceding two, an infinite sum. I will often use the **Sigma-Summation notation**, which uses the upper-case Greek letter Sigma, Σ , and indices like we saw with indexed set operations.

von Neumann Construction of $\mathbb{N}$

Hungarian mathematician, physicist, and the computer Scientist who designed modern computer architecture, János "Johnny" von Neumann submitted a paper in his teen years with the following construction of $\mathbb{N}$ as the first step in his construction of the set of transfinite ordinals. I don't care what the hell they identify as, I won't extend our treatment to them here in this fine house of the lord. In all seriousness, $\mathbb{N}$ is all we need for right now.

Johnny the Martian (read Labatut's The MANIAC), like any good Computer Scientist, starts off his set of counting numbers with zero, represented by the empty set $\emptyset = \{\}$. The Successor Function is defined as $n+1 = n \cup \{n\}$. Since the empty set has no elements to union with $\{\emptyset\}$, $1 = \{\emptyset\}$. Continuing with this rule we see

$$2 = 1 \cup \{1\} = \{\emptyset, \{\emptyset\}\},$$

$3 = 2 \cup \{2\} = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$, and generally we can see that by building $\mathbb{N}$ this way, a given natural number's cardinality is equal to that very number.

Relations, functions, mapping

One Set operation we seemingly skipped right over earlier is the **Cartesian product**, named after 17th Century French polymath René Descartes. Descartes was a pioneer in connecting geometry to algebra, he allowed line segments of length one and demonstrated how to find monstrous polynomials in the analytic study of curved lines. An n -ary Cartesian product is the set of length- n lists that can be made from n -many sets, with the i^{th} element of a given list coming from the i^{th} set. Symbolically we denote a cartesian product like multiplication with a $\times$, $\cdot$, or, if we're taking the cartesian product of the same set p times, we can throw a superscript p above said set like exponentiation.

Suppose we're given sets $A = \{a_1, a_2, a_3, \dots, a_n\}$ and $B = \{b_1, b_2, b_3, \dots, b_m\}$, then $A \times B = \{(a, b) : a \in A, b \in B\}$.

A **relation** is a subset of the cartesian product of sets



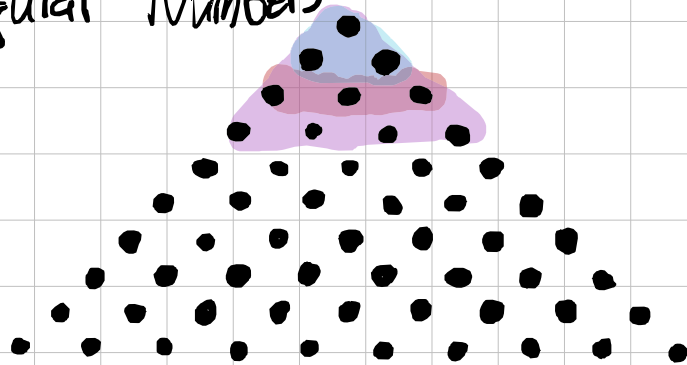
Problems

Bibliography

Figurate Out

All that work we've just done was a student-led introduction to some of the publicly lesser-known tools and definitions of those foolish mathematicians, as well as the construction of our first "stage" or "playing field." But who's going to act in our plays written in logic on a paper-thin stage? What games can be played and who plays them, and in what ways? Honestly if you can describe a thing, you can write math about or with it, sometimes that description is the math that we want to do about our given thing. So that I may make clear both the beauty and power

Triangular Numbers



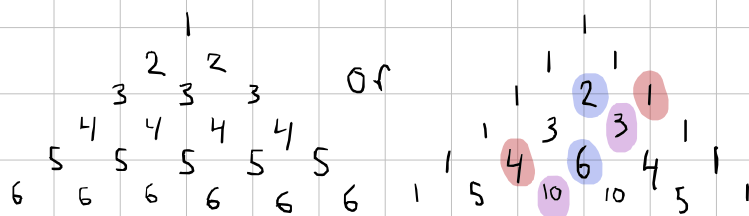
The triangular numbers T_n count the number of objects it takes to form an equilateral triangle of side-length n . Above is a visual representation of T_n for $n=10$. Notice that as we increment n by 1, the number we add to the sequence also increases by 1.

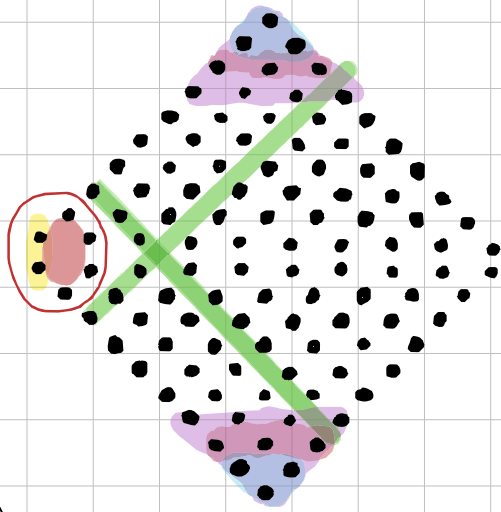
$$n = 1, 2, 3, 4, 5, 6, \dots$$

$$T_n = 1, 3, 6, 10, 15, 21, \dots$$

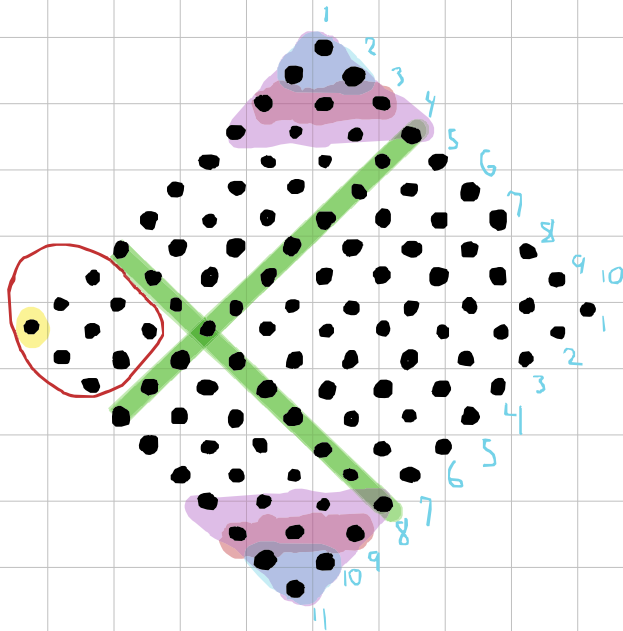
The apparent pattern is $T_n = \frac{n(n+1)}{2}$

We'll find this to be the case by pondering the triangle. Recall from geometry that the area for a triangle is given by $\frac{1}{2} \cdot \text{base} \cdot \text{height}$, which we'll recall we can prove by copying a given triangle & creating a rectangle from them.





We do appear to have a square of side length n , which would yield $\frac{n^2}{2}$, but notice how the dot-lattice is disturbed and there's also an extra pair of sides born from the One-to-One side alignment. We'll fix by sliding one of the rectangles a unit over.



It's now obvious by the graphic that we can follow through with the proof. Recounting the Sides after our corrective shift shows that for a triangle of side $n=10$, T_n is indeed in the form $\frac{n(n+1)}{2}$. More importantly than proving the specific case, we've shown it necessary for finding the "area" in dots of an equidistantly arranged, equilateral triangle (definition of triangular numbers), we must use a rectangle of "Area" $n(n+1)$.

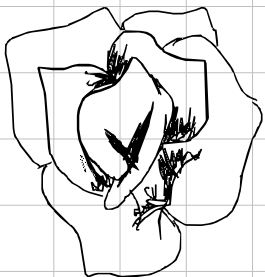
We can see that $T_n = 1+2+3+\dots+(n-1)+n$ by analyzing the triangles on previous pages, but now we'll prove it by connecting that Series Sum to our recently proven formula $T_n = \frac{n(n+1)}{2}$.

Say $S = 1+2+3+\dots+(n-2)+(n-1)+n$. If we throw a duplicate in reverse order underneath we'll have

$$\begin{aligned} S &= 1 + 2 + 3 + \dots + (n-2) + (n-1) + n \\ S &= n + (n-1) + (n-2) + \dots + 3 + 2 + 1 \end{aligned}$$

Now we have two S 's on the left hand side + n -many pairs of $(n+1)$, in mathematical notation:

$2S = n(n+1)$, dividing both sides by two gives $S = \frac{n(n+1)}{2}$, therefore we've now shown our formula for T_n to be equal to the sum of natural numbers from 1 to n . ■



Since we've brought up sums, let's review the Sigma-Summation notation. In sigma sum notation, we use a capitalized Greek letter sigma, Σ , to denote that what follows will be a sum of values. We typically index sums expressed this way by choosing a variable, setting it equal to the sum's lower-bound, and then writing the upper bound on top

Therefore the statement we've just proven can be expressed as:

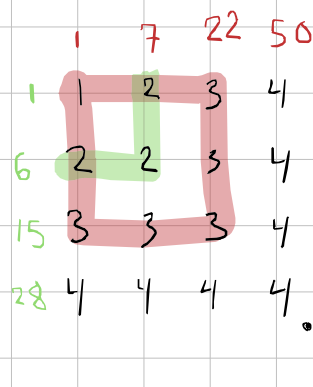
$$T_n = \sum_{x=0}^n x$$

Square Numbers

The next shape we can build should be very familiar

• • • •
• • • •, etc. The SQUARES!!

We can recall the square of the n^{th} number is denoted n^2 and is given by $n \cdot n$. This follows by the definition of a square as a special rectangle whose width is equal to its height.



To prove this, let's take a look at the dot in the opposite corner from our "Seed Square" (1). This dot, no matter the size of n or n^2 , is an element of both sets of sides we've added n to. To fix this we subtract it once from the twice we've added it to the square, thus each new square is given by adding the same function of n , $2n-1$ (n^{th} odd counting 1).

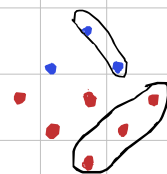
In the previous argument, we basically proved

Reader Exercises 1

Draw dot diagrams for T_3 , T_4 , and T_5

Draw dot diagrams for 3^2 , 4^2 , and 5^2

Use dots to show that $T_n + T_{n-1} = n^2$



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